esa ECOSPHERE

Guidelines for constructing allometric models for the prediction of woody biomass: How many individuals to harvest?

S. H. Roxburgh,[1,†] K. I. Paul,[1] D. Clifford,[2] J. R. England,[3] and R. J. Raison[1]

[1]*Commonwealth Scientific and Industrial Research Organisation Agriculture Flagship and Land and Water Flagship, GPO Box 1700, Canberra ACT 2601 Australia*
[2]*Commonwealth Scientific and Industrial Research Organisation Digital Productivity Flagship, GPO Box 2583, Brisbane, Queensland 4001 Australia*
[3]*Commonwealth Scientific and Industrial Research Organisation Agriculture Flagship and Land and Water Flagship, Private Bag 10, Clayton South, Victoria 3169 Australia*

Citation: Roxburgh, S. H., K. I. Paul, D. Clifford, J. R. England, and R. J. Raison. 2015. Guidelines for constructing allometric models for the prediction of woody biomass: how many individuals to harvest? Ecosphere 6(3):38. http://dx.doi.org/10.1890/ES14-00251.1

Abstract. The recent development of biomass markets and carbon trading has led to increasing interest in obtaining accurate estimates of woody biomass production. Aboveground woody biomass (B) is often estimated indirectly using allometric models, where representative individuals are harvested and weighed, and regression analyses used to generalise the relationship between individual mass and more readily measured non-destructive attributes such as plant height and stem diameter (D). To satisfy regulatory requirements and/or to provide market confidence, allometric models must be based on sufficient data to ensure predictions are accurate, whilst at the same time being practically and financially achievable.

Using computer resampling experiments and allometric models of the form $B = aD^b$ the trade-off between increasing the sample size of individuals to construct an allometric model and the accuracy of the resulting biomass predictions was assessed. A range of algorithms for selecting individuals across the stem diameter size-class range were also explored. The results showed marked variability across allometric models in the required number of individuals to satisfy a given level of precision. A range of 17–95 individuals were required to achieve biomass predictions with a standard deviation within 5% of the mean for the best performing stem diameter selection algorithm, while 25–166 individuals were required for the poorest. This variability arises from (a) inherent uncertainty in the relationship between diameter and biomass across allometric models, and (b) differences between the diameter size-class distribution of individuals used to construct a model, and the diameter size-class distribution of the population to which the model is applied.

Allometric models are a key component of quantifying land-based sequestration activities, but despite their importance little attention has been given to ensuring the methods used in their development will yield sufficiently accurate biomass predictions. The results from this study address this gap and will be of use in guiding the development of new allometric models; in assessing the suitability of existing allometric models; and in facilitating the estimation of uncertainty in biomass predictions.

Key words: afforestation; allometry; precision; reforestation; shrub biomass; tree biomass.

Received 5 August 2014; revised 15 November 2014; accepted 2 December 2014; final version received 8 January 2015; **published** 26 March 2015. Corresponding Editor: J. Nippert.

Copyright: © 2015 Roxburgh et al. This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. http://creativecommons.org/licenses/by/3.0/

† **E-mail:** stephen.roxburgh@csiro.au

Introduction

Management of the world's forest ecosystems has the potential to significantly contribute to climate change protection through enhancing carbon sequestration in new and existing forests, through the judicious management of forest products including bioenergy substitution, and through avoided deforestation and degradation (Canadell and Raupach 2008, Jose and Bardhan 2012). The recent establishment of biomass and carbon markets has provided a mechanism for these goals to be achieved (van Oosterzee 2012, Macintosh 2013), with the potential to deliver additional natural resource management co-benefits such as the restoration of degraded ecosystems and associated improvements to biodiversity (Taiyab 2005, Chazdon 2008, Lin et al. 2013, Polglase et al. 2013).

Participation in biomass or carbon markets requires estimating changes in vegetation biomass through time, with biomass estimated with sufficient accuracy to satisfy the minimum standards specified in the appropriate legislative instruments, and/or to ensure market confidence. The most accurate method for estimating biomass is to directly harvest and weigh individual trees or shrubs. However due to cost constraints, direct harvest methods are impractical to implement at large spatial scales, and they are destructive. More commonly, allometric models that relate direct measurements of the aboveground biomass of a sub-sample of the population to non-destructive measurements, such as stem diameter and plant height, are used for prediction. Using such allometric relationships, non-destructive measurement inventories can be undertaken over large areas and biomass estimated, albeit with an inherent level of uncertainty. There exists a very large literature on the development and use of allometric models for estimating forest biomass, reviewed by Picard et al. (2012), and documented in the GlobAllometree database (http://www.globallometree.org/).

Once established, allometric models can be a cost effective and efficient method for accurate biomass estimation. However, their initial development and validation can be expensive as it typically involves sample selection, harvesting, measurement of plant height and stem diameters, weighing, and sub-sampling of plant components (e.g., crown and bole) for laboratory-based moisture determination to convert fresh weight to dry weight (Brown 2002, Picard et al. 2012). This cost could significantly impact on the economic viability of activities, especially if numerous individuals are required. Surprisingly, relatively little attention has been given to determining optimal sample sizes required to establish allometric relationships that yield biomass predictions with a given level of accuracy. Indeed, we are only aware of three such studies (Cunia 1986*a*, Chave et al. 2004, van Breugel et al. 2011). We are not aware of any studies that have specifically investigated the optimal diameter size-class distribution of trees to construct allometric models, which also impacts on the accuracy of biomass estimates.

A wide range of non-destructive measurements have been used as independent variables for predicting tree biomass, with the most common measures being stem diameter and height, but other attributes such as wood density and indices of canopy volume have also been employed. An equally diverse array of functional models has been used to relate these measurements to biomass, with a corresponding broad range of methods for estimating model parameters (Picard et al. 2012). Here we focus on simple power functions that relate stem diameter to biomass (Equation 1), as these have been shown to adequately capture the allometry of a broad range of species (Kuyah et al. 2012, Paul et al. 2013*b*, *c*). The power function can be linearized by logarithmic transformation (Eq. 2), thus allowing parameter estimates and model uncertainty to be estimated using standard linear algebra:

$$B = aD^b \tag{1}$$

$$\ln(B) = \ln(a) + b \times \ln(D) + \varepsilon. \tag{2}$$

Logarithmic transformation also corrects allometric data for heteroscedasticity. Residual errors on the logarithmic scale (ε) are assumed to be normally distributed with zero mean and constant variance. In this study aboveground biomass (B) is given in units of kg/individual, and diameters (D) are in cm.

The aims of this study are twofold. First, to use published allometric relationships to explore both the implications for biomass prediction of

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

varying the sample size (i.e., number of individuals used to construct the relationship), and the importance of the diameter size-class distribution of that sample. Second, to use these results to develop guidelines for the construction and use of allometric models for revegetation and carbon sequestration projects, with a particular focus on ensuring adequate standards of accuracy can be met within the constraints of destructive field harvesting programs.

Methods

Overview

The approach follows that of Chave et al. (2004) and van Breugel et al. (2011), whereby an existing allometric relationship, based on a sample of n_p harvested individuals, is used as the basis for computer resampling experiments. Statistics from this existing relationship are used in a Monte-Carlo procedure, with replicate samples of n_s individuals of varying diameter drawn at random from the observed relationship, and a new allometric model constructed for each replicate. These sampled allometric models are then applied to field-inventory measurements to predict site-based biomass. For a given sample size n_s, the degree of repeatability of biomass predictions across replicate samples is used to quantify the precision of the estimate, measured as the coefficient of variation in predicted biomass, CV_B:

$$CV_B = s_B/\bar{x} \tag{3}$$

where $\bar{x}$ and s_B are the mean and standard deviation of predicted biomass across the replicate samples, respectively. High values of CV_B indicate that the biomass prediction for a single allometric relationship constructed from n_s individuals could be either appreciably higher or lower than the true value, simply due to inter-sample variability. As n_s increases, replicate biomass predictions become more similar, with a corresponding increase in the confidence in the reliability of the allometric model.

The precision of the biomass prediction, CV_B, quantifies one component of overall accuracy. The second component is bias, which is the tendency to either consistently over- or underestimate the true population value. There are two sources of potential bias during allometric model development. Firstly, bias can arise due to non-random measurement errors during harvesting, e.g., poorly calibrated weighing equipment. Strict field sampling protocols were adhered to during the harvesting of trees used in this study, thus this potential source of bias is considered negligible. Secondly, the non-random selection of individuals, e.g., subjective choice of trees for harvesting by field personnel, can lead to model bias. The allometric models included in this study were all developed adhering to standard statistical practice to ensure randomness and representativeness during the selection of individuals for harvest, and hence the samples can be considered unbiased. Because these sources of bias are negligible, the overall accuracy of model prediction due to varying allometric model sample size is therefore dominated by errors in precision, CV_B.

Bias can also occur during the application of the resulting allometric model if the population being estimated is not well represented by the model. This can arise in a single-species allometric model when, for example, for a given diameter the individuals that are being estimated have either consistently more or less biomass than the average biomass predicted by the model. Bias can also arise similarly in 'generic' models that combine many species if a target species is consistently over- or underestimated by the model (van Breugel et al. 2011). Although these sources of bias are potentially significant, they are associated with the subsequent application of the allometric model (as opposed to its development), and therefore their assessment lies beyond the scope of the current analysis.

Allometric datasets

Computer resampling experiments were based on 23 published allometric models for species that are commonly planted in southeast Australia for biomass or carbon markets. Nineteen of these models were species-specific, classified into tree-form *Eucalyptus* (11 species) and *Acacia* species (3 species), mallee-form *Eucalyptus* species (a life form that branches from ground level; 3 species), and *Acacia* shrubs (2 species) (Table 1). These 19 species are a subset of those reported in Paul et al. (2013*b*, *c*). These 23 allometric models were selected based on the requirement that they had at least 50 harvested individuals. In addition,

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Summary of 23 allometric models used in the Monte-Carlo analyses, with number of individuals used to construct the model (n_p), height of stem diameter measurement (H), mean and range of diameter measurements, allometric model parameters (a and b), and fit statistics (R^2 and residual standard deviation, s).

Allometric model	n_p	H (cm)	Stem diameter range (cm)	Mean diameter (cm)	Slope (a)	Intercept (b)	R^2	s
Mallee-form Eucalypts								
E. lox-liss†	1077	10	1.2–22.9	8.9	2.59	−2.86	0.94	0.30
E. polybractea	882	10	1.1–34.8	10.7	2.51	−2.81	0.94	0.35
E. kochii	368	10	1.1–27.0	11.6	2.43	−2.83	0.94	0.37
Eucalypt trees								
E. wandoo	54	130	2.3–27.2	9.0	2.38	−2.18	0.94	0.39
E. viminalis	365	130	2.3–29.8	10.0	2.30	−2.19	0.95	0.24
E. tricarpa	54	130	3.6–54.2	19.5	2.40	−2.39	0.95	0.31
E. tereticornis	71	130	2.6–46.9	12.5	2.34	−2.15	0.96	0.25
E. loxophleba	56	130	2.9–29.0	10.1	2.23	−1.74	0.95	0.25
E. spathulata	206	130	2.3–41.3	9.7	2.22	−1.30	0.95	0.25
E. polyanthermos	51	130	1.8–20.1	7.9	2.05	−1.46	0.96	0.26
E. occidentalis	118	130	2.3–79.0	11.7	2.43	−2.12	0.98	0.24
E. melliodora	169	130	1.8–39.2	7.7	2.12	−1.73	0.94	0.33
E. largiflorens	57	130	1.6–18.0	9.8	2.01	−1.23	0.94	0.20
E. camaldulensis	88	130	1.7–63.0	13.7	2.32	−1.97	0.98	0.27
Acacia trees and shrubs								
A. calamifolia	128	10	1.0–15.9	6.0	2.41	−2.23	0.94	0.36
A. hakeoides	110	10	1.0–21.3	7.0	2.18	−2.26	0.94	0.35
A. acuminata	110	130	1.1–27.6	5.2	2.13	−1.65	0.92	0.46
A. melanoxylon	51	130	1.0–23.4	8.2	2.15	−1.70	0.98	0.26
A. saligna	50	130	1.9–39.8	9.5	2.42	−2.23	0.95	0.37
Generalized allometric models								
Tree	2332	130	2.5–101.1	12.3	2.27	−1.83	0.91	0.36
Acacia shrub	499	10	2.0–32.2	7.9	2.24	−2.23	0.85	0.45
Non-*Acacia* shrub	249	10	2.7–33.3	8.4	2.08	−2.45	0.83	0.41
Mallee *Eucalyptus*	2313	10	1.1–34.8	10.0	2.49	−2.74	0.93	0.35

† *Eucalyptus loxophleba* ssp. *lissophloia*.

four generalised allometric models, comprising aggregations of many individual species, were also investigated (Generic *Acacia* shrubs, Generic non-*Acacia* shrubs, Generic mallee-form *Eucalyptus*, and Generic tree) (Paul et al. 2013*b*, *c*). Full details on the field harvest methods and statistical fitting procedures for these allometric models are given in Paul et al. (2013*a*, *b*, c). Although these allometric models are dominated by the genera *Eucalyptus* and *Acacia* they represent a wide range of tree morphologies. Indeed, the morphological diversity within the data, as expressed by the range of model parameters and model fit statistics, is comparable with the range of parameter values and variability found globally for models of the same form as Eq. 1 (Appendix A).

The performance of simulated allometric models was assessed using independent field measurements of stem diameter, collected from a range of sites across southern Australia (Table 2). For each allometric model, data from the field site having the highest number of measured individuals was used as the basis for testing (Paul et al. 2013*a*, *d*). The decision to use a single site for assessing performance was because the availability of sites with sufficient inventory records to make a meaningful test was, for many species, limited to a single site. The exception to this was the Generic mallee-form *Eucalyptus* allometric model. Because mallee plantings are predominantly monocultures, predictions from this model were tested using field inventory data aggregated across multiple sites in order to include measurements from all three representative mallee species.

The site-level total biomass prediction (in t ha^{-1}) corresponding to each allometric model was calculated by iterating through the field diameter measurements, and summing, over all individuals, the allometric-predicted back-transformed biomass values. Variability in total biomass between replicate sampling experiments was then used as the estimate of CV_B. When

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Summary of the field inventory data on which the allometric models were tested. Full site descriptions can be found in Paul et al. (2013*a*, *b*, c), and the data is published in Paul et al. (2013*d*).

Site name	Allometric model tested	*n*	*D* (cm)
Mallee-form Eucalypts			
Pepal	*E. lox-liss*†	774	7.6
Kalawa	*E. polybractea*	2169	7.2
McFall	*E.kochii*	94	8.8
Eucalypt trees			
McFall	*E. wandoo*	585	8.1
Strathearn	*E. viminalis*	327	9.6
Leos	*E. tricarpa*	18	14.3
Netherleigh	*E. tereticornis*	1585	12.4
Moir	*E. loxophleba*	41	7.8
Moir	*E. spathulata*	206	9.7
Strathearn	*E. polyanthermos*	40	7.2
Gunbower	*E. occidentalis*	255	19.2
Strathearn	*E. melliodora*	145	6.2
Gunbower	*E. largiflorens*	255	11.8
Gunbower	*E. camaldulensis*	248	7.3
Acacia trees and shrubs			
Jenharwill	*A. calamifolia*	122	5.7
Jenharwill	*A. hakeoides*	87	6.2
McFall	*A. acuminata*	156	6.8
Lynvale	*A. melanoxylon*	1049	4.4
McFall	*A. saligna*	154	7.2
Generalized allometric models			
Netherleigh	Generic tree	2804	10.5
Palomar	Generic *Acacia* shrub	285	6.0
Suttons	Generic non-*Acacia* shrub	635	7.1
Various	Generic mallee *Eucalyptus*	9796	6.3

† *Eucalyptus loxophleba* ssp. *lissophloia*.

calculating biomass from log-log models, care must be taken to avoid bias when back-transforming from the logarithmic to the natural scale. To remove this bias a number of correction factors have been proposed (Picard et al. 2012). In this study the 'MM' correction factor (Shen and Zhu 2008) was used, as described and recommended in the review by Clifford et al. (2013).

Simulation design

There are three key components that determine the overall uncertainty of a given allometric model when it is used to estimate biomass, and these form the basis of the simulation experiments. First, there is the inherent variability in the population under consideration, reflecting the natural range of variability of individual tree or shrub biomass for a given diameter. On the natural scale, this variability increases with increasing diameter (Fig. 1A), but can be assumed constant on the logarithmic scale (Fig. 1B). This variability is summarised as the standard deviation of residuals (on a logarithmic scale) around the line of best fit, s. (Fig. 1B, Table 1). The second source of uncertainty contributing to the precision of biomass estimates is the number of individuals on which the allometric model is based, with precision increasing with increasing sample size n_s (sampling uncertainty). The final source of uncertainty is the interplay between the size-class distribution of stem diameters used in the construction of the allometric model, and the size-class distribution of stem diameters to which the model is applied (Cochran 1977, Picard et al. 2012). This is a key aspect, because regression-based estimates of biomass are most precise close to the mean of the stem diameter used to construct the allometric model, and become progressively less precise away from that mean (Fig. 2). Therefore, applying an allometric model to a population with a different underlying diameter size-class distribution to that used to construct the model will result in reduced precision (Cochran 1977, Snowdon et al. 2002).

Selection of individuals for developing allometric relationships must therefore be carefully

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

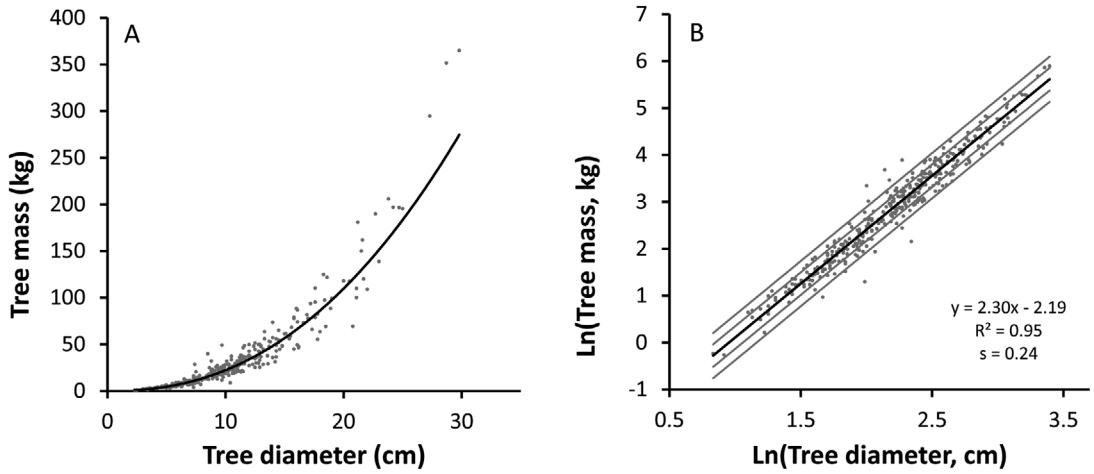

Fig. 1. Relationship between tree diameter and measured tree mass for *Eucalyptus viminalis* ($n = 365$) on (A) the natural scale, and (B) the logarithmic scale. Parallel lines in (B) are the line of best fit (black), and ±1 and ±2 SD (s) (grey).

considered, as the natural variability in the population, the total number of individuals harvested, and their size-class distribution all impact on how accurate the ultimate biomass predictions will be. Manual harvesting is also both expensive and time-consuming, and therefore it is important to elucidate how these key components interact. This was investigated via six scenarios, described below.

For each scenario, the core of the analysis involved constructing new allometric models, under various constraints, by sampling at random from the observed allometric relationship. To construct a new allometric model, n_s diameters were selected from a pre-defined size-class distribution (see Scenarios 1–6 below), and for each diameter the Ln(biomass) value was drawn from a normal distribution with a mean equal to the expected value given by the linear parameters a and b (Eq. 2), and standard deviation s. Note this differs from the resampling procedure adopted by Chave et al. (2004) and van Breugel et al. (2011), whose analyses were based on bootstrapped re-sampling from the actual data. The parametric sampling adopted here was deemed necessary, as in many cases the sample sizes were insufficient in the observed data to ensure an even and representative spread of observations around the fitted line. For example, with small sample sizes there are often a small (<5) number of individuals representing the largest size class. These individuals can, by chance, fall asymmetrically around the line of best fit. Bootstrapped resampling of the observed data repeatedly selects these 'unrepresentative' individuals, leading to biased biomass predictions for these larger size classes. Parametric resampling therefore ensures an unbiased selection of individuals across the whole diameter size class domain.

Each of the six scenarios was applied to the 23 allometric models listed in Table 1, thus covering a range of natural variability (s). For all scenarios n_s was varied from 5 to 1000, in increments of 5, and the relationship between sample size and precision summarised by plotting n_s against CV_B (Chave et al. 2004). For all scenarios, individuals were selected from within the observed diameter range for each of the 23 allometric models (Table 1). For each sample size (n_s), values of CV_B were calculated from 1000 randomly constructed models. The six scenarios are differentiated by the algorithm used to select individuals across the range of observed diameters, i.e., the diameter size-class distribution.

Scenario 1. Individuals selected to match the size-class distribution of inventory trees.—Under this scenario the size-class distribution of individuals selected to construct the allometric model is specifically matched to the size-class distribution of individuals in the target population to which the model is intended to be applied. Because we

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

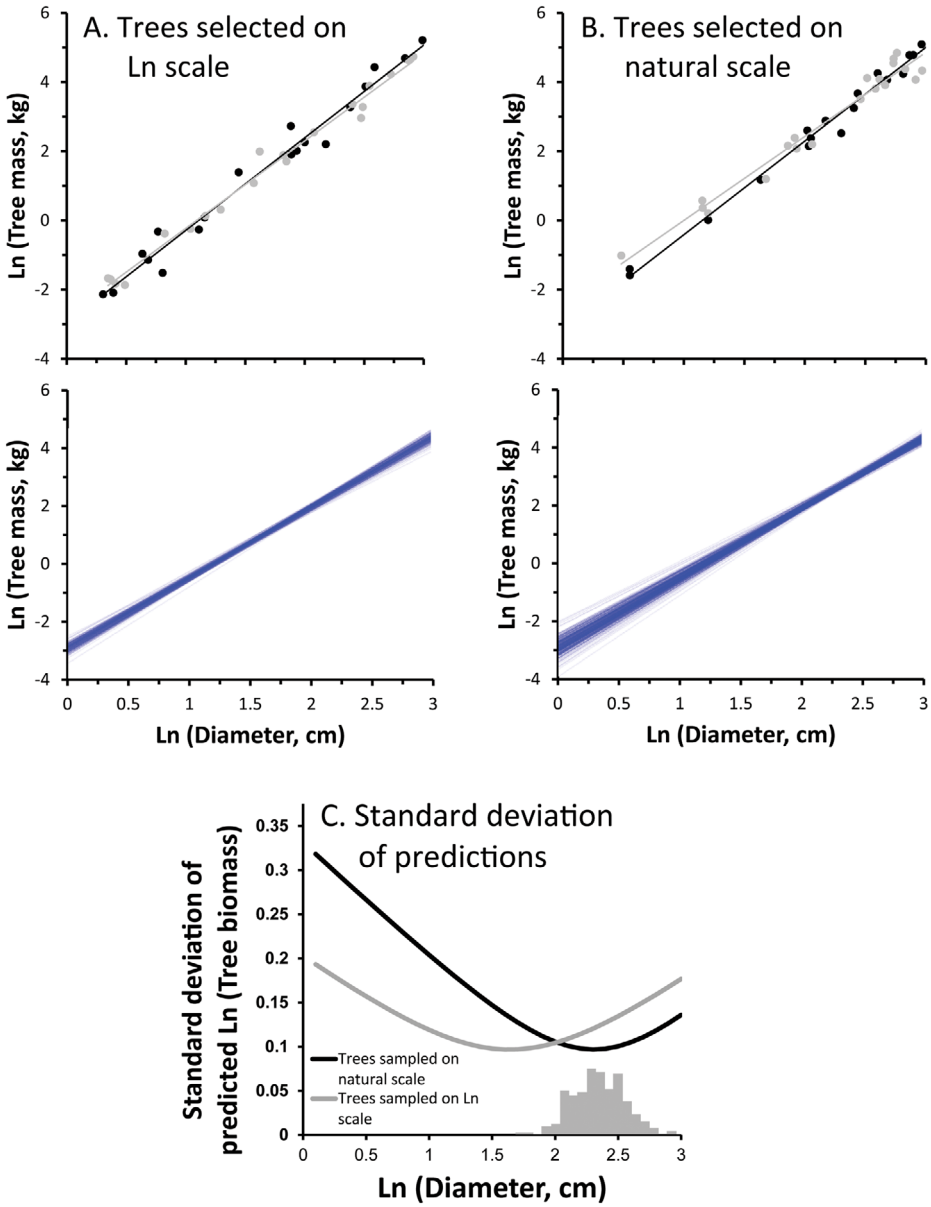

Fig. 2. Illustration of between-sample variability in fitted regression lines as a function of tree selection method. The top panel of (A) shows two sets each of 20 individuals (grey and black data points) selected at random and uniformly on the ln(D) scale, and in (B) individuals are selected at random and uniformly on the natural scale (see also Fig. 3C, E). The lower panels of (A) and (B) show fitted regression lines for 500 realisations, illustrating that variability in predicted ln(biomass) is lowest around the around the mean of the ln(D) of trees used to construct the allometric model, which differs between selection methods. In (C) the standard deviation of predicted ln(biomass) is plotted against ln(D) for each selection method. The frequency histogram shows a hypothetical size distribution of field inventory data, suggesting for this example that an allometric model constructed with individuals selected evenly on the natural scale (B) will yield ln(biomass) predictions which are less variable than allometric models constructed using method (A).

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

are concerned with obtaining the most precise estimate of biomass on the natural scale (and not the logarithmic scale on which the model is constructed), the selection of individuals in this scenario is biased towards those diameter size classes that are expected to contribute most to the total site biomass. In many cases this is the relatively smaller number of large individuals within the stand (Brown and Lugo 1992, Brown et al. 1997, Chave et al. 2005, Roxburgh et al. 2006, Dean et al. 2012).

Biomass is approximately proportional to the square of the diameter, D^2 (Snowdon et al. 2002, Zianis and Mencuccini 2004, Chave et al. 2005, Fehrmann and Kleinn 2006). Therefore, random diameters were selected from the cumulative frequency distribution of field-measured D^2, using the method described in Snowdon et al. (2002) and with the cumulative sum of D^2 divided into 10 equal strata, with equal probability sampling within each stratum (Fig. 3A). The decision to use 10 strata on which to base the probability of selection is arbitrary. Based on a limited survey, Cochran (1977) suggested six strata as being optimal. The sensitivity of our results to the number of strata was investigated by repeating all analyses with 6 and 20 strata. Minimal differences were found between the runs, hence only the results for 10 strata are presented.

In contrast to Scenario 1, Scenarios 2–6 apply allometric models to field populations of inventory trees that may have differing diameter size-class distributions to those used in the construction of the models.

Scenario 2. Individuals selected to match the size-class distribution of diameters used to construct the original allometric model.—Scenario 2 replicates the common situation where an existing allometric model is applied to a new site, without necessarily considering the size-class distribution of the diameters at that site. Here, individuals are selected only to match the size-class distribution of the individuals used in the construction of the original allometric model. To implement this algorithm a similar selection procedure to that described in Scenario 1 was used, but using the cumulative frequency distribution of Ln(D) of the individuals used to construct the original allometric model, again with 10 strata (Fig. 3B).

Scenarios 3–6 give consideration to the construction of new allometric models that seek to broadly represent the range of diameters to which it is expected it will be applied. For example, Dietz and Kuyah (2011) recommend first identifying the likely diameter range of application, and then dividing that range into 5–6 equal strata, with random selection of an equal number of individuals per stratum. An example of the implementation of this strategy is Montagu et al. (2005). This approach has also been specified in carbon sequestration legislation (CFI 2013).

Scenario 3. Random sample on the natural scale.—In Scenario 3, n_s individuals are selected from a uniform distribution in the range $D_{\min}$ to $D_{\max}$. This leads to a distribution of individuals on the logarithmic scale that is biased towards the larger size classes (Fig. 3C).

Scenario 4. Stratified random sample on the natural scale (5 strata).—This is analogous to the recommendation of Dietz and Kuyah (2011), where the range $D_{\min}$ to $D_{\max}$ is split into five equal strata, with an equal number of individuals selected at random (on a uniform distribution) within each stratum (Fig. 3D).

Scenario 5. Random sample on the logarithmic scale.—This is similar to Scenario 3, except individuals are selected uniform on the logarithmic scale in the range Ln($D_{\min}$) to Ln($D_{\max}$). This leads to an even distribution (and variability) on the logarithmic scale, but on the natural scale is biased towards the smaller size classes (Fig. 3E). Because the statistical fitting is done on the logarithmic scale, this scenario generates allometric models with overall residual variation that is minimised, and that is symmetric around the mean Ln(D) (compare Fig. 2A, B).

Scenario 6. Stratified random sample on the logarithmic scale (5 strata).—For this scenario the range Ln($D_{\min}$) to Ln($D_{\max}$) is split into five equal strata, with an equal number of individuals selected at random (on a uniform distribution) per stratum (Fig. 3F).

For all scenarios the relationship between CV_B and n_s is used to explore the effect of increasing sample size on the precision of predicted biomass. This relationship follows a power function, with precision increasing in proportion to $\sqrt{n_s}$ (Picard et al. 2012). For each scenario, and for each allometric model, the required sample size to achieve a given level of precision was

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

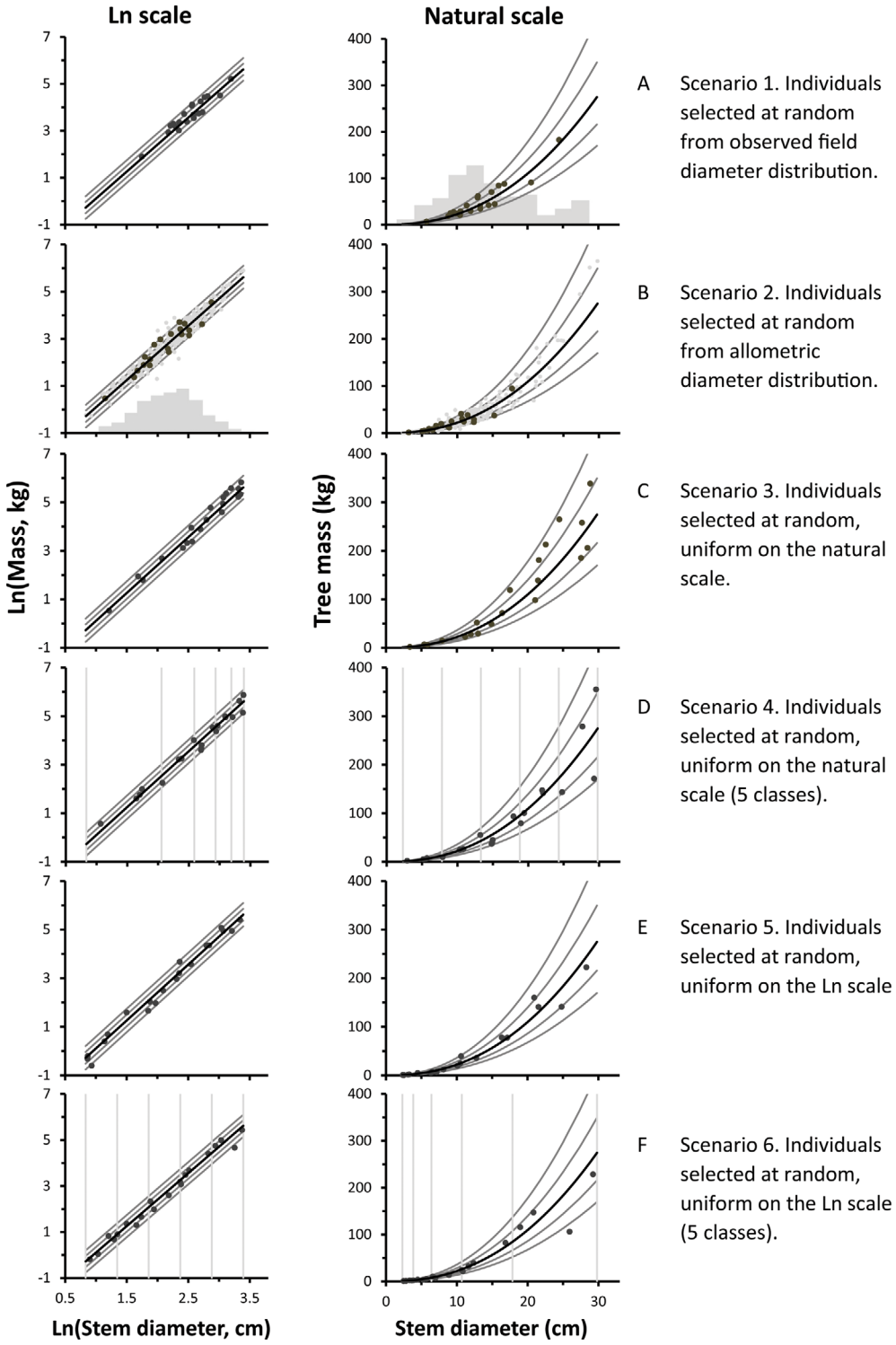

Fig. 3. Graphical representation of the six scenarios (1–6) used to select individuals across a range of diameters for constructing random allometric relationships. The error bars (grey) are ±1 and ±2 SD (*s*). See *Methods* for more details.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

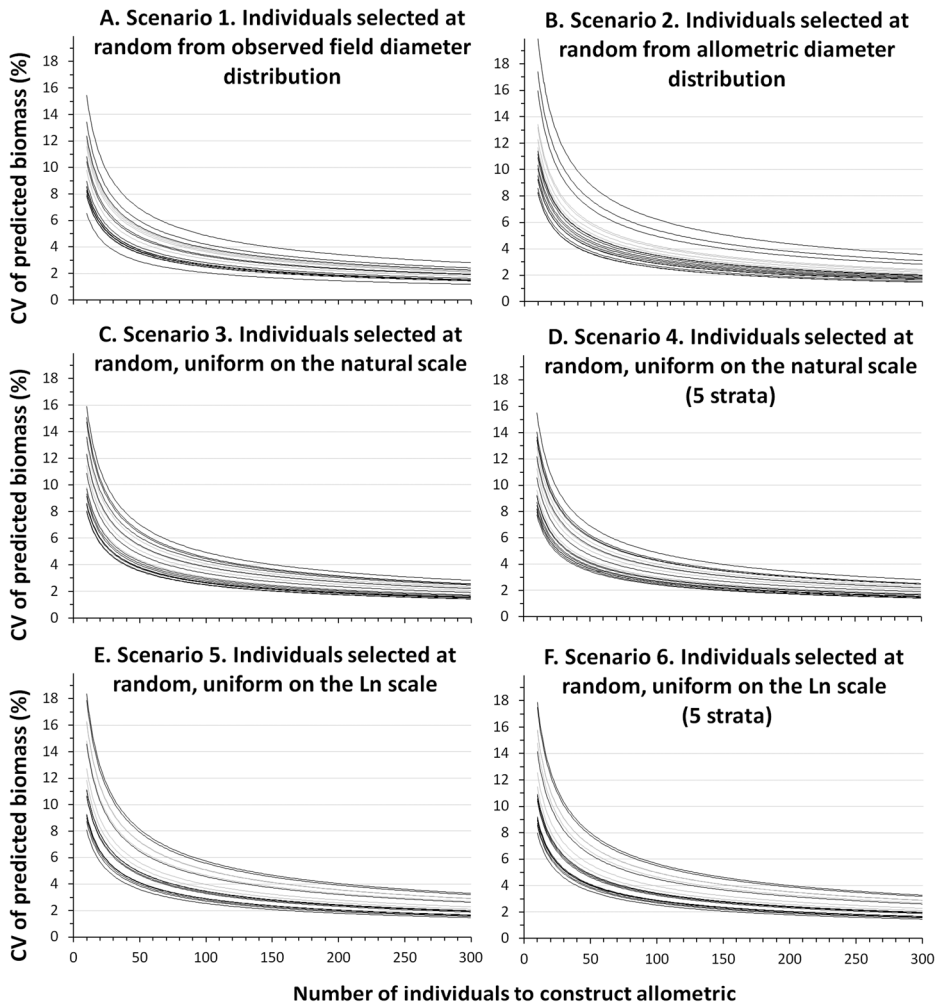

Fig. 4. Precision in predicted biomass (CV_B) vs. the sample size of individuals used to construct the species-specific allometric models (n_s) for each of the six stem diameter selection scenarios (1–6). Black lines are *Eucalyptus* and *Acacia* tree species, and grey lines are *Acacia* shrubs and mallee-form *Eucalyptus*.

estimated via interpolation by fitting an empirical power function to the relationship between calculated CV_B and n_s, with the sample size (n_α) required to achieve a given level of precision (CV_α) calculated as:

$$n_\alpha = e^{\frac{\ln(CV_\alpha/a')}{b'}} \quad (4)$$

where a' and b' are fitted parameters. Sample sizes for three levels of precision are reported in the results ($CV_{0.05}$, $CV_{0.075}$, and $CV_{0.10}$), corresponding to biomass predictions with standard deviations that are 5%, 7.5% and 10% of the mean, respectively. Additional results for $CV_{0.01}$, and $CV_{0.025}$ are also provided in Appendix B.

Results

Individual allometric model sampling curves

The power function relationship between the coefficient of variation of predicted biomass (CV_B) and the sample size used to construct the allometric model (n_s) shows an initial steep decline with increasing sample size that starts to level off beyond approximately 50 individuals (Figs. 4 and 5). Notably, all six scenarios are characterised by a high level of between-model variability in the number of individuals required to achieve a given level of precision.

The results of the analysis are discussed

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

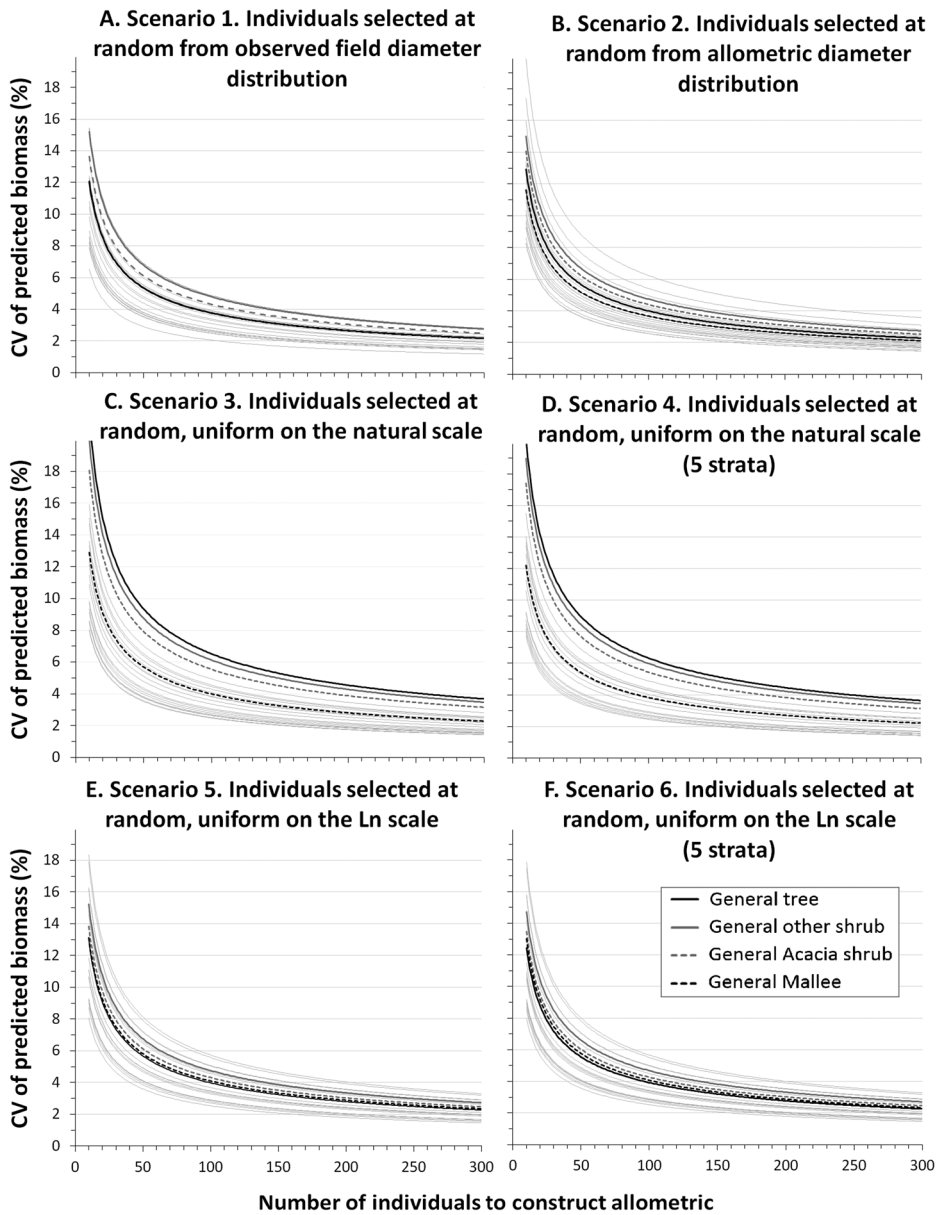

Fig. 5. Precision in predicted biomass (CV_B) vs. the sample size of individuals used to construct the generic allometric models (n_s) for each of the six stem diameter selection scenarios (A–F). Grey lines are included for reference, and correspond to the species-specific allometric results (Fig. 4).

primarily with respect to achieving a 5% CV_B (although results with CV_B targets of 7.5% and 10% are also presented) A CV_B target of 5% is the same level of precision adopted by Chave et al. (2004), and approximately equates to estimating biomass with an error of ±10% at the 95% level of confidence (IPCC 2003). It also strikes an appropriate balance between constraints on the numbers of individuals that can be practically harvested, and satisfying current legislative requirements (CFI 2013).

For the species-specific allometric models (Fig. 4), the sample size required to achieve a 5% CV_B for Scenario 1 ranged from 17 (*Eucalyptus largiflorens*) up to 95 (*Acacia acuminata*) (Fig. 4A). When compared with the scenario yielding

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the lowest overall precision (Scenario 5; Fig. 4E) the equivalent sample sizes ranged from 26 (*E. loxophleba*) up to 131 (*A. acuminata*). Similar trends were also apparent with the generalised allometric models (Fig. 5), albeit with overall lower precision due to the greater inherent variability that occurs when data from many species are combined. Over all six scenarios the minimum required sample size was 54 (Scenario 2, Generic mallee allometric model; Fig. 5B), and the maximum 166 (Scenario 3, Generic tree allometric model; Fig. 5C).

When selecting individuals on either the natural or logarithmic scale there was very little difference in the required sample sizes between simple random selection of individuals within the diameter range, and a five-class stratified random selection across the same diameter range (e.g., compare Fig. 4C, D). Indeed, for a given level of precision, the average difference between required sample sizes for random and stratified sampling was less than two individuals (Appendix B). Although stratification, in general, leads to improved precision of sampled estimates, in our simulations the number of individuals associated with the precision range of interest (CV_B = 5–10%) is large enough that simple random sampling is sufficient to ensure a repeatable and representative spread of diameter values. Despite this, stratification remains a sound basis on which to select samples and is often recommended when selecting individuals for harvest (Dietz and Kuyah 2011). Therefore in the aggregated results discussed below, only Scenarios 4 and 6 (i.e., those based on stratified sampling) are considered, in addition to Scenarios 1 and 2.

Aggregated results

Although the relationship between n_s and CV_B for the species-specific allometric models (Fig. 4) and for the generalised allometric models (Fig. 5) illustrates the underlying power function relationship between sample size and precision, differences among the scenarios can be more clearly seen when the required sample sizes to achieve specific levels of precision (CV_α = 5%, 7.5% and 10%) are calculated from these curves and aggregated into the major species groups (Fig. 6). For the species-specific allometric models, Scenario 1 (Fig. 6A) yielded the most precise predictions with an average of 56, 51 and 40 individuals (to achieve $CV_B = 5\%$) for the *Acacia* shrubs, mallee-form *Eucalyptus*, and tree-form *Eucalyptus* and *Acacia* groups respectively. Selecting individuals from the diameter range on the natural scale (Scenario 4; Fig. 6C) yielded similar results to Scenario 1, with 54, 53 and 47 individuals, respectively. Scenario 2 (Fig. 6B), which explored precision in predicted biomass when an existing allometric model is applied to a new population, was significantly worse for the *Acacia* shrubs and tree-form *Eucalyptus* and *Acacia* groups (70 and 56 individuals, respectively), but similar to the other two scenarios for the mallee-form *Eucalyptus* (50 individuals). The least precise estimates occurred when individuals were selected at random on the log scale (Scenario 6; Fig. 6D), with values of 93, 72 and 52, respectively.

For the generalised allometric models, Scenario 1 also yielded the most precise predictions, with a required 91, 74, 57 and 57 individuals to achieve a 5% CV for Generic *Acacia* shrubs, Generic non-*Acacia* shrubs, Generic mallee-form *Eucalyptus*, and Generic tree allometric models, respectively (Fig. 6A). However, the pattern for the remaining scenarios differed from the species specific models. In the case of the generalised allometric models Scenarios 2 (Fig. 6B) and 6 (Fig. 6D) yielded results similar to Scenario 1, with 88, 73, 67 and 62 individuals respectively for Scenario 6, and 90, 77, 54 and 64 individuals for Scenario 2. In contrast to the species-specific allometric models where selecting individuals on the natural scale (Scenario 4; Fig. 6C) produced relatively precise predictions, predictions for the generalised allometric models for this scenario were appreciably worse, yielding 143, 118, 62 and 160 individuals, respectively.

Required sample sizes for values of $CV_\alpha = 7.5\%$ and $CV_\alpha = 10\%$ were consistently 0.45 and 0.25 of the values for CV_α = 5.0%, respectively. This conversion can be applied to all scenarios, and a full tabulation of all results is given in Appendix B.

Generalization

Displaying the curves shown in Figs. 4 and 5 on the logarithmic scale reveals an underlying power relationship of the form:

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

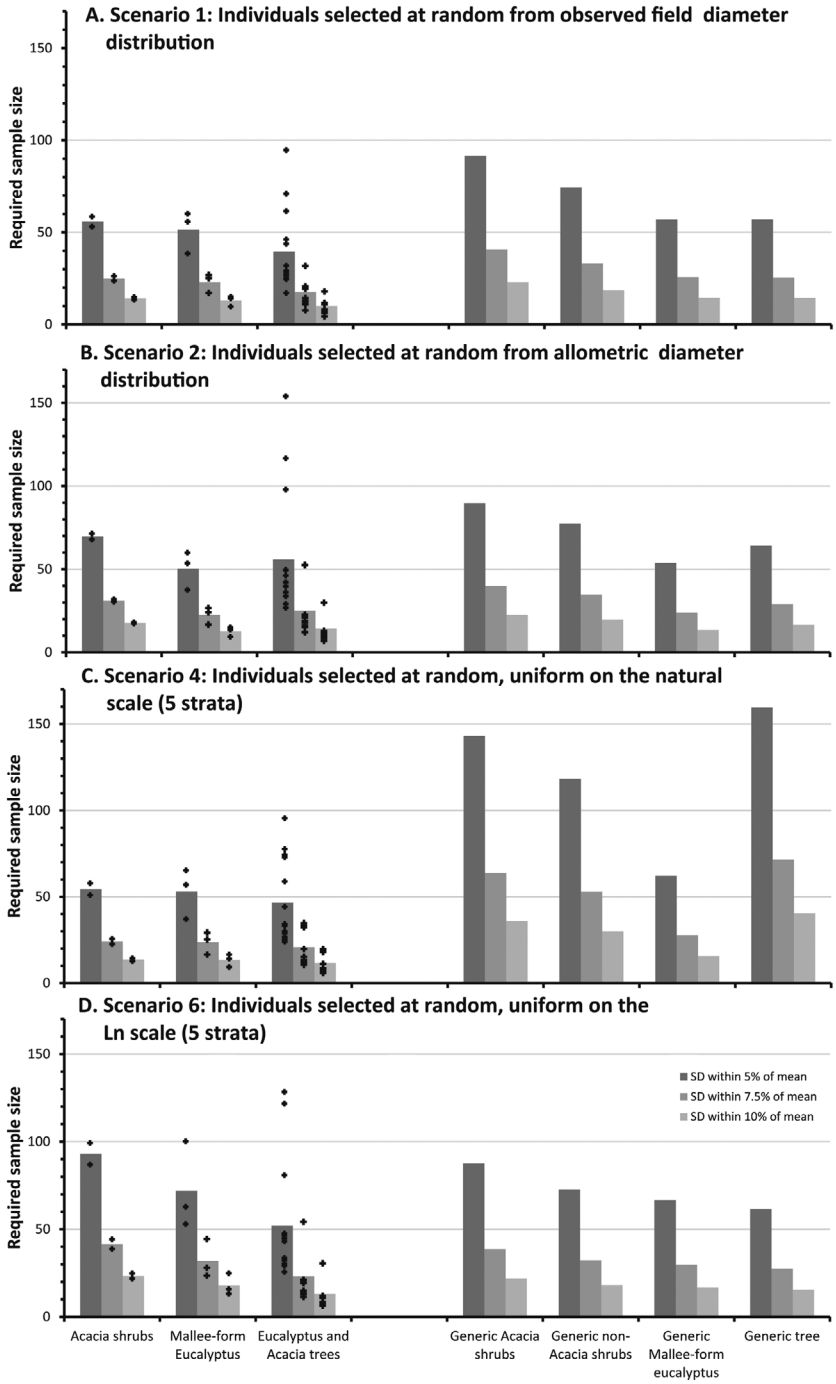

Fig. 6. Required sample sizes (number of individuals to construct allometric models) to satisfy three levels of precision (biomass predicted with coefficients of variation (CV_B) of 5%, 7.5% and 10%), for each of the three major species groupings (*Acacia* trees and shrubs, mallee-form eucalypts and Eucalypt trees) and for each of the four generic allometric models. Crosses show the results for each individual species for the species-specific allometric models. Due to similarity in response only four of the six tree selection scenarios are shown (Scenarios 1, 2, 4, 6, panels A–D, respectively). See text for further details.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

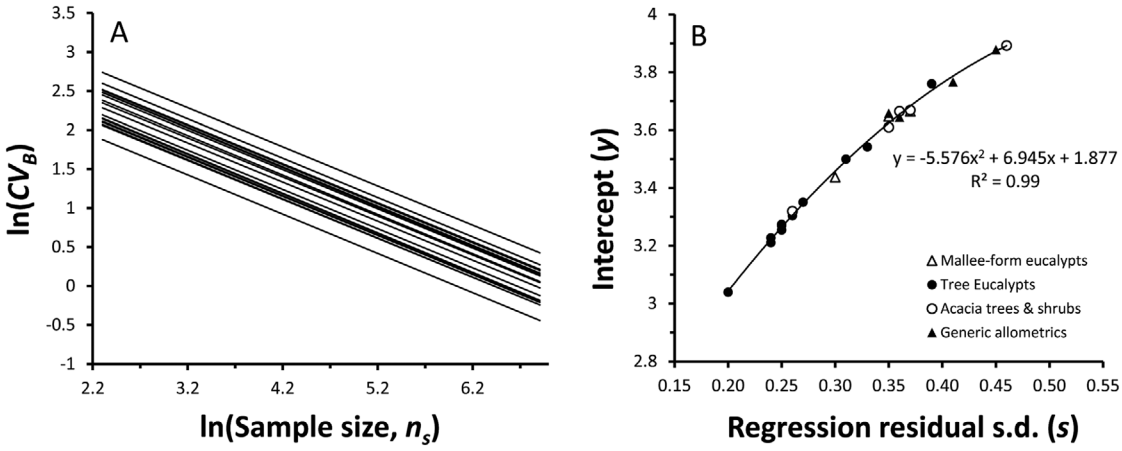

Fig. 7. When expressed on the logarithmic scale the relationship between predicted biomass precision (CV_B) and sample size (n_s) reveals a power function, with a common exponent of −0.5 (indicating precision increases as a function of $\sqrt{n_s}$). Panel A shows this relationship for all 23 allometric models, with individuals selected according to Scenario 1 (i.e., with trees selected to ensure congruence with the diameter size-class distribution of the field population to which the model was applied). Under this scenario, between-allometric variation in estimated precision (the differences in the intercepts of the lines in panel A) is explained by the variability of residuals in the parent allometric model (s, Table 1). This is shown in panel B, where the y-intercepts from panel A are plotted against s, with the relationship approximated by a second-order polynomial.

$$\ln(\mathrm{CV}_B) = c \times \ln(n_S) + d \tag{5}$$

with a series of parallel lines with $c = -0.5$, and intercepts (d) that are functions of the variability associated with each allometric model (Fig. 7A), which as noted in the methods are functions of (1) the inherent variability around the line of best fit, s, and (2) the difference in size-class distributions between the individuals used to develop the allometric model, and the population to which the model is applied. In general, d will be difficult to predict given it is a joint function of these two components, the latter of which will be case-specific and dependent upon the characteristics of the range of stands to which the allometric model is applied. However in the case of Scenario 1, where uncertainty due to (2) has been minimised by ensuring commonality in the diameter size-class distributions, then there is a very tight relationship between the intercept d, and the remaining source of variability, s (Fig. 7B). This predictability of d can be used to generalise the results for this scenario, yielding a simple equation that can be used to either estimate the expected precision for an existing allometric model (given s and n_p), or if a new allometric model is to be created, provide an estimate of the required sample size n_p to achieve a given level of precision for a given s. Because any additional difference in the size-class distributions between the allometric diameters and the field diameters results in an increase in CV_B for a given n_s, this generalization thus provides minimum estimates for these quantities for any given allometric model.

The relationship between precision and sample size for any single allometric model (Fig. 7A) is given by:

$$\ln(\mathrm{CV}_B) = -0.5 \times \ln(n_S) + d \tag{6}$$

$$\mathrm{CV}_B = \frac{e^d}{\sqrt{n_S}} \tag{7}$$

where d can be approximated by the relationship in Fig. 7B:

$$d = -5.576s^2 + 6.945s + 1.877. \tag{8}$$

Through substitution, the expected precision for an existing allometric model with known s and n_s can thus be computed by

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$$\mathrm{CV}_B = \frac{e^{-5.576s^2+6.945s+1.877}}{\sqrt{n_S}} \tag{9}$$

and the required sample size to achieve a given level of precision if s can be computed by

$$n_S = \left(\frac{e^{-5.576s^2+6.945s+1.877}}{\mathrm{CV}_B}\right)^2. \tag{10}$$

For convenience, tabulated values for these quantities for a range of scenarios are given in Appendix C.

Discussion

Given the central importance of allometric models for predicting biomass in both native and planted systems, it is surprising the relationship between the methods used in their construction and their ultimate performance in predicting biomass have been largely overlooked. Apart from an earlier simulation study report in Cunia (1986*a*), the only previous studies of which we are aware are that of Chave et al. (2004) and van Breugel et al. (2011), who both investigated, as part of broader analyses of error propagation in tropical forests, the effect of sample size on the precision of biomass predictions.

Our analysis identified three major sources of variability contributing to uncertainty in allometric biomass predictions. The first has been reported previously (Chave et al. 2004, van Breugel et al. 2011), and is the increase in precision that accompanies an increase in the sample size (n_s) of individuals used to construct the allometric model. This is a general statistical phenomenon, with precision increasing as a function of $\sqrt{n_s}$ (Picard et al. 2012). Because this source of variability can be readily manipulated, it was selected as the primary axis on which our results are presented.

The second source of variability is the inherent population level-variability in the relationship between stem diameter and biomass. For a given allometric relationship, the observed variability around a fitted regression line (s) combines both a sampling error component (which is a function of n_s) and the underlying natural variability. As n_s increases towards the total number of individuals in the population, the sampling error component approaches zero, at which point s describes the population-level variability. In the datasets analyzed here (Table 1), a minimum sample size of $n = 50$ was stipulated to ensure the allometric models on which the analyses were based provide a reasonable approximation of this population level variability, where $n = 50$ corresponds to estimating the true population-level variability with a CV of approximately 10%.

The final source of variability is one that has received relatively little attention, and relates to the difference in the size-class distributions of individuals used to construct the allometric model, and the size-class distribution of individuals to which the allometric model is applied. This is potentially a major source of uncertainty, given the simulations predicted up to 100% variation in required sample sizes to achieve the same level of precision across different diameter size-class selection algorithms (Fig. 6). The major implication is that uncertainty in predicted biomass is not solely a function of the characteristics of the allometric model being applied, but is also a function of the characteristics of the population which is being estimated. Overall, the most precise biomass estimates are obtained when the size frequency distribution used in the construction of the allometric model matches that of the population to which it is applied. Indeed, this is the approach advocated by Cunia (1986*b*) where an initial non-destructive inventory of tree diameters is used to quantify the population size-class structure, and this information is then used to inform the selection of trees to construct the model. An alternative approach is to select trees to span more evenly the potential diameter range to which the model is expected to be applied, such as that advocated by Dietz and Kuyah (2011). More commonly, descriptions of (and justifications for) the selection of individuals to construct allometric models are not reported by researchers (van Breugel et al. 2011).

When the size-class distribution of individuals selected to construct the allometric model matches that of the population under consideration (Scenario 1), uncertainty due to differences in the size-class distributions is minimised, and between-allometric differences in the relationship between sample size and biomass precision are well predicted by s alone (Fig. 7). This leads to a general mathematical relationship (Eq. 9) that

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

can be used to provide estimates of precision for a given combination of sample size (n_s) and allometric model variability (s), or alternatively can be used to estimate the required sample size to achieve a given level of precision. Because Scenario 1 generates the most precise predictions, the estimates from Eq. 9 should be considered best-case scenarios, yielding minimum sample sizes, and maximum attainable precision. For situations where it is expected the size-class distributions differ between the allometric model and the population, then these estimates will need to be adjusted accordingly.

Scenario 2, which explores the precision of biomass predictions when existing allometric relationships are applied to field data, showed similar trends to Scenario 1, albeit with lower precision. This suggests the ranges of diameters that underlie the observed allometric models used in this study are broadly representative of the range of diameters present in the field data. For the scenarios where individuals were selected to represent an even range of diameters on the natural (Scenarios 3 and 4) or logarithmic scales (Scenarios 5 and 6), there were differences in the patterns of required sample sizes between the species-specific and the generalised allometric models. The generalised allometric models produced the greatest precision when individuals were selected evenly on the logarithmic scale, whereas the species-specific allometric models performed better when individuals were selected evenly on the natural scale. This difference is again a reflection of the field diameter size-class distributions to which the allometric models were applied, with the Generic *Acacia* shrub, Generic non-*Acacia* shrub and Generic tree test sites tending to have a higher proportion of smaller individuals compared with the species-specific allometric models, thus resulting in a more even spread of values on the logarithmic scale, and hence greater precision. The notable exception was the Generic mallee-form *Eucalyptus* allometric model which yielded relatively precise predictions when individuals were selected evenly on the natural scale. This is because mallee plantings are typically established by tube stock rather than by seed, resulting in relatively low spatial variability in stand density (Paul et al. 2013*c*), and relatively low variability in individual stem diameters.

When species were grouped into major life-form classes, the species-specific allometric models tended to yield more precise predictions compared to the generalised models, due in part to overall lower population-level variability around the fitted regression line (s). There was, however, large between-species variability in required sample sizes within each major life-form class (Fig. 6). Across the major species groupings, the shrub life form, which is generally less regular in shape and thus exhibits greater variability in biomass for a given stem diameter, required a correspondingly greater sample size to achieve the same levels of precision compared to more regular tree-form species. Mallee-form trees, which are multi-stemmed and typically branch from ground level, showed levels of precision that were intermediate between the shrubs and the more traditional tree-form species.

The results of our simulations were summarised at three levels of precision, corresponding to biomass predictions with standard deviations that are 5%, 7.5% and 10% of the mean, but with a focus on 5%. In practice, the definition of a 'sufficiently' precise estimate is a subjective one, and depends on the context with which the allometric model is intended to be used; the level of precision required for an ecological study may be quite different to that required for predicting biomass for commercial harvesting operations, or where monetary payments are being made based on estimates of biomass or carbon sequestration (van Breugel et al. 2011).

Because the development of allometric relationships are within our control and represent only one part of the chain of uncertainty when biomass is estimated at the total stand scale, it seems prudent to strive towards minimising the error associated with the allometric prediction component as much as practicable. In recommending appropriate sample sizes for developing generic tropical tree allometric models, Chave et al. (2004) adopted $CV_B = 5\%$ as an acceptable limit, corresponding in their analysis to a required minimum sample size of 100 trees. This is similar to the required sample sizes for the Generalised tree allometric model considered in this study, with required sample sizes, at the same level of precision, ranging from 57 to 160 depending on the stem diameter selection algo-

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

rithm used (Fig. 6).

It should be noted however, that analyses of allometric model uncertainty presented here represent only one component of a range of different sources of variability that contribute to the total uncertainty of stand-level biomass, and which additionally includes individual tree measurement error, error associated with field sampling design, and the potential for prediction bias during the application of allometric models (Cunia 1986*b*, Parresol 1999, Chave et al. 2004, van Breugel et al. 2011). For example, Chave et al. (2004) concluded that error from allometrics alone was approximately 50% of the total uncertainty for biomass predictions in a 5-ha tropical forest plot.

Recommendations

Whenever possible, tree and shrub selection protocols for constructing allometric models should be guided by the size-class distribution of diameters in the population to which the model is intended to be used. For generalized allometric models constructed under these conditions, our analyses suggest minimum sample sizes of 55 individuals for Generic tree and Generic mallee-form *Eucalyptus*, and 75–100 individuals for Generic *Acacia* and non-*Acacia* shrubs (Table 3). Required sample sizes for species-specific allometric models can be either significantly higher or lower than these values, depending on the inherent variability of the species in question. In the absence of a priori information on species variability, and using our average simulation results as a guide, harvesting a minimum of 40 individuals for regular tree-form species, and 50 individuals for mallee-form species is recommended. The *Acacia* shrubs category contains only two species therefore generalizations must be treated with caution; results based on this limited sample suggest 50–60 individuals as appropriate.

Constructing new allometric models is not always possible, nor practical, and it is common to apply existing equations to new field sites. For these situations, the above sample sizes may need to be increased to compensate for the additional uncertainty associated with any differences in the size-class distributions between the individuals used to develop the model, and the sample population. Our simulation results from Scenario 2 can be used as a guide but are not definitive, as they are based on limited testing using available field inventory data, and therefore will be sensitive to the particular size-class characteristics of those sites. With that in mind, compared with the Scenario 1 results, the Scenario 2 simulations suggest little difference in recommended sample sizes for the species-specific mallee-form *Eucalyptus*, Generic mallee-form *Eucalyptus*, Generic *Acacia*, and Generic non-*Acacia* allometric models (Table 3). In contrast, the species-specific *Acacia* shrubs and tree-form *Eucalyptus* and *Acacia* categories required a 27% and 38% increase in sample-size respectively, and the Generic tree allometric model required an 18% increase in sample size (Table 3). On a species-by-species basis, 7 of 19 species required an additional 50–70% increase in sample size to maintain the same level of precision when moving from Scenario 1 to Scenario 2 (Appendix B). Because this variability in response across species and allometric categories is due to site-specific differences in the respective diameter-size class distributions, general recommendations are difficult to make. What is clear is that differences in the variability of tree sizes within the population that an allometric model is to be applied, relative to the variability of tree sizes used to construct the model, can have an important implications for the resulting biomass predictions. This should be taken into account wherever possible, e.g., whenever allometric models have been constructed with individuals sampled evenly on the natural or logarithmic scales (Scenarios 2–6; Montagu et al. 2005, Dietz and Kuyah 2011, CFI 2013).

Overall our results suggest many existing site-and-species specific allometric models may be based on insufficient sample numbers, given predictions based on models with sample sizes of less than 20–30 individuals may subject to high uncertainty (even though individual allometric models may have high statistical significance and sound R^2 metrics and model efficiencies). The fact that large sample sizes are required for site-and-species specific allometric models raises questions regarding the cost effectiveness of utilising these for estimating biomass at larger estate or regional scales. In contrast, generic equations outperform site-and-species specific equations in terms of area of

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Recommended sample sizes to achieve biomass predictions with a standard deviation within 5% of the mean ($CV_B = 5\%$) for Scenarios 1 and 2. The ranges for the species-specific allometric models are shown in parentheses. *n* indicates the number of species within each species-specific life-form category. The change in required sample size reflects the site-specific effects of differences in diameter size-class distributions between each allometric model and the field site data against which the model is applied.

Allometric model type	Diameter size-class selection algorithm†		Change in required sample size (%)
	Scenario 1	Scenario 2	
Species-specific			
Eucalyptus and *Acacia* trees ($n = 14$)	40 (17–95)	55 (27–154)	37.5
Mallee-form *Eucalyptus* ($n = 3$)	50 (38–60)	50 (38–60)	0
Acacia shrubs ($n = 2$)	55 (53–59)	70 (68–71)	27.2
Generalized			
Generic tree	55	65	18.2
Generic mallee-form *Eucalyptus*	55	55	0
Generic *Acacia* shrub	75	75	0
Generic non-*Acacia* shrub	100	100	0

† Scenario 1: Individuals selected at random to match observed field diameter distribution; Scenario 2: Individuals selected at random to match allometric diameter distribution.

applicability and general utility, and in many cases will be more cost effective whilst still delivering acceptable levels of precision (Paul et al. 2013*a*, *b*).

The above recommendations have associated with them a number of caveats. First, it was assumed that all field measurements will fall within the diameter range of the individuals used to construct the allometric model. As noted by many authors, extrapolating beyond the bounds of the model potentially introduces significant bias and additional uncertainty (Picard et al. 2012). Second, analyses were based on a simple power function, and do not include additional terms such as tree height, D^2, wood density, or additional parameters that account for changing allometry as individuals increase in size (Parresol 1999, Chave et al. 2004), and that could lead to increased precision. When applied to our datasets, the simple power model provided robust allometric relationships, with previous work concluding that the inclusion of other factors such as height or wood density would be unlikely to yield significant improvements over the simpler model (Paul et al. 2013*b*). However for other datasets the inclusion of additional variables might be required. The degree to which additional parameters would improve the overall precision of biomass estimates will be case-sensitive and thus difficult to generalise. There is also a trade-off to consider between the increased cost and complexity of making additional measurements, and the cost of simply increasing the sample size and retaining the simpler model. These are important considerations given the need to provide cost-effective yet accurate methods for biomass estimation. Exploration of these various options remains a topic for future work.

Finally, it was assumed there are no errors associated with either the field measurement of stem diameters, nor in the estimation of biomass. If the total uncertainty of stand-level biomass is of interest, then more extensive analyses involving error propagation across different scales of measurement are required, including errors associated with sample design and the spatial distribution of sample plots, tree measurement error, and errors associated with the use of allometric models (Chave et al. 2004, Yanai et al. 2010, Nickless et al. 2011, Ahmed et al. 2013).

Acknowledgments

This work was supported in part by a grant from the Australian Government through The Department of the Environment. For discussions during the early phases of this work we thank A. O. Nicholls. For providing constructive criticism on the manuscript we thank A. O. Nicholls, G. Roberts, C. Dean and two anonymous reviewers.

Literature Cited

Ahmed, R., P. Siqueira, S. Hensley, and K. Bergen.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2013. Uncertainty of forest biomass estimates in north temperate forests due to allometry: implications for remote sensing. Remote Sensing 5:3007–3036.

Basuki, T. M., P. E. van Laake, A. K. Skidmore, and Y. A. Hussin. 2009. Allometric equations for estimating the above-ground biomass in tropical lowland *Dipterocarp* forests. Forest Ecology and Management 257:1684–1694.

Brown, S. 2002. Measuring carbon in forests: current status and future challenges. Environmental Pollution 116:363–372.

Brown, S., and A. E. Lugo. 1992. Aboveground biomass estimates for tropical moist forests of the Brazilian Amazon. Interciencia 17:8–18.

Brown, S., P. Schroeder, and R. Birdsey. 1997. Aboveground biomass distribution of US eastern hardwood forests and the use of large trees as an indicator of forest development. Forest Ecology and Management 96:37–47.

Canadell, J., and M. Raupach. 2008. Managing forests for climate change mitigation. Science 320:1456–1457.

CFI. 2013. Carbon credits (Carbon Farming Initiative) (Reforestation and Afforestation) methodology determination 2013. Federal Register of Legislative Instruments F2013L01210. Canberra, Australia.

Chave, J., et al. 2005. Tree allometry and improved estimation of carbon stocks and balance in tropical forests. Oecologia 145:87–99.

Chave, J., R. Condit, S. Aguilar, A. Hernandez, S. Lao, and R. Perez. 2004. Error propagation and scaling for tropical forest biomass estimates. Philosophical Transactions of the Royal Society B 359:409–420.

Chazdon, R. L. 2008. Beyond deforestation: Restoring forests and ecosystem services on degraded lands. Science 320:1458–1460.

Clifford, D., N. Cressie, J. R. England, S. H. Roxburgh, and K. I. Paul. 2013. Correction factors for unbiased, efficient estimation and prediction of biomass from log-log allometric models. Forest Ecology and Management 310:375–381.

Cochran, W. G. 1977. Sampling techniques. Third edition. John Wiley & Sons, New York, New York, USA.

Cunia, T. 1986*a*. Evaluating errors of tree biomass regressions by simulation. Pages 235–242 *in* Proceedings of the Workshop on Tree Biomass Regression Functions and their Contribution to the Error of Forest Inventory Estimates. USDA Forest Service, USA.

Cunia, T. 1986*b*. Error of forest inventory estimates: its main components. Pages 1–13 *in* Proceedings of the Workshop on Tree Biomass Regression Functions and their Contribution to the Error of Forest Inventory Estimates. USDA, Forest Service, USA.

Dean, C., N. B. Fitzgerald, and G. W. Wardell-Johnson. 2012. Pre-logging carbon accounts in old-growth forests, via allometry: an example of mixed-forests in Tasmania, Australia. Plant Biosystems 146:223–236.

Dietz, J., and S. Kuyah. 2011. Guidelines for establishing regional allometric equations for biomass estimation through destructive sampling. World Agroforestry Centre (ICRAF), Nairobi.

Eamus, D., K. McGuinness, and W. Burrows. 2000. Review of allometric relationships for estimating woody biomass in Queensland, the Northern territory and Western Australia. National Carbon Accounting System Technical Report 5a. Commonwealth of Australia, Canberra, Australia.

Fehrmann, L., and C. Kleinn. 2006. General considerations about the use of allometric equations for biomass estimation on the example of Norway spruce in central Europe. Forest Ecology and Management 236:412–421.

IPCC. 2003. Good practice guidance for land use, land-use change and forestry. The Institute for Global Environmental Strategies (IGES), Japan.

Jose, S., and S. Bardhan. 2012. Agroforestry for biomass production and carbon sequestration: an overview. Agroforestry Systems 86:105–111.

Keith, H., D. Barrett, and R. Keenan. 2000. Review of allometric relationships for estimating woody biomass for New South Wales, the Australian Capital Territory, Victoria, Tasmania and South Australia. National Carbon Accounting System Technical Report 5b. Commonwealth of Australia, Canberra, Australia.

Kuyah, S., J. Dietz, C. Muthuri, R. Jamnadass, P. Mwangi, R. Coe, and H. Neufeldt. 2012. Allometric equations for estimating biomass in agricultural landscapes: I. Aboveground biomass. Agriculture Ecosystems & Environment 158:216–224.

Lambert, M. C., C. H. Ung, and F. Raulier. 2005. Canadian national tree aboveground biomass equations. Canadian Journal of Forest Research 35:1996–2018.

Lin, B. B., S. Macfadyen, A. Renwick, S. A. Cunningham, and N. Schellhorn. 2013. Maximizing the environmental benefits of carbon farming through ecosystem service delivery. BioScience 63:793–803.

Macintosh, A. 2013. The Carbon Farming Initiative: removing the obstacles to its success. Carbon Management 4:185–202.

Montagu, K., K. Duttmer, C. Barton, and A. Cowie. 2005. Developing general allometric relationships for regional estimates of carbon sequestration: an example using *Eucalyptus pilularis* from seven contrasting sites. Forest Ecology and Management 204:113–127.

Nickless, A., R. Scholes, and S. Archibald. 2011. A method for calculating the variance and confidence intervals for tree biomass estimates obtained from

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

allometric equations. South African Journal of Science 107:86–95.

Parresol, B. 1999. Assessing tree and stand biomass: A review with examples and critical comparisons. Forest Science 45:573–593.

Paul, K., et al. 2013*a*. Improved estimation of biomass accumulation by environmental planting and mallee plantings using FullCAM. Report for The Department of the Environment. CSIRO Sustainable Agriculture Flagship, Canberra, Australia.

Paul, K. I., et al. 2013*b*. Development and testing of allometric equations for estimating above-ground biomass of mixed-species environmental plantings. Forest Ecology and Management 310:483–494.

Paul, K. I., et al. 2013*c*. Testing allometric equations for prediction of above-ground biomass of mallee eucalypts in southern Australia. Forest Ecology and Management. 310:1005–1015.

Paul, K. I., et al. 2013*d*. Mallee Eucalypt plantings inventory. Version 5. CSIRO. Data Collection. 10.4225/08/527262C031E58. https://data.csiro.au/dap/browse

Picard, N., S.-A. Laurent, and M. Henry. 2012. Manual for building tree volume and biomass allometric equations: from field measurement to prediction. Food and Agricultural Organization of the United Nations, Rome and Centre de Cooperation Internationale en recherche Agronomique pur le Developpement, Montpellier, France.

Polglase, P. J., A. Reeson, C. S. Hawkins, K. I. Paul, A. W. Siggins, J. Turner, D. F. Crawford, T. Jovanovic, T. J. Hobbs, K. Opie, J. Carwardine, and A. Almeida. 2013. Potential for forest carbon plantings to offset greenhouse emissions in Australia: economics and constraints to implementation. Climatic Change 2:161–175.

Roxburgh, S. H., S. W. Wood, B. G. Mackey, G. Woldendorp, and P. Gibbons. 2006. Assessing the carbon sequestration potential of managed forests: a case study from temperate Australia. Journal of Applied Ecology 43:1149–1159.

Shen, H. P., and Z. Y. Zhu. 2008. Efficient mean estimation in log-normal linear models. Journal of Statistical Planning and Inference 138:552–567.

Snowdon, P., D. Eamus, P. Gibbons, P. Khanna, K. H. J. Raison, and M. Kirschbaum. 2000. Synthesis of Allometrics, review of root biomass and design of future woody biomass sampling strategies. National Carbon Accounting System Technical Report 17. Commonwealth of Australia, Canberra, Australia.

Snowdon, P., R. J. Raison, H. Keith, P. Ritson, P. Grierson, M. Adams, K. Montagu, H. Bi, W. Burrows, and D. Eamus. 2002. Protocol for sampling tree and stand biomass. National Carbon Accounting System Technical Report No. 31. The Australian Greenhouse Office, Canberra, Australia.

Taiyab, N. 2005. The market for voluntary carbon offsets: A new tool for sustainable development? IIED, London, UK.

van Breugel, M., J. Ransijn, D. Craven, F. Bongers, and J. Hall. 2011. Estimating carbon stock in secondary forests: Decisions and uncertainties associated with allometric biomass models. Forest Ecology and Management 262:1648–1657.

van Oosterzee, P. 2012. The integration of biodiversity and climate change: A contextual assessment of the carbon farming initiative. Ecological Management & Restoration 13:238–244.

Yanai, R., J. Battles, A. Richardson, C. Blodgett, D. Wood, and E. Rastetter. 2010. Estimating uncertainty in ecosystem budget calculations. Ecosystems 13:239–248.

Zianis, D., and M. Mencuccini. 2004. On simplifying allometric analyses of forest biomass. Forest Ecology and Management 187:311–332.

Supplemental Material

Appendix A

Comparison of allometric models with global datasets

The 19 species-specific and four generic allometric equations included in the main analysis are dominated by the genera *Acacia* and *Eucalyptus*. Although the range of morphological variability represented within these 23 equations is high, there is interest in comparing how this variability compares with a broader range of species, to gauge more generally the representativeness of the equations on which the main analyses are based. This was achieved by

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

compiling a global database of allometric models for aboveground biomass that are of the same form as Eq. 1.

Methods

A selection of 189 equations were compiled from the published literature and from the GlobAllomeTree database (http://www.globallometree.org/). The global data includes 36 equations for temperate and boreal Canadian trees from the review of Lambert et al. (2005) (using the above-ground 'wood' component model parameters); 31 equations for tropical forest trees from the review of van Breugel et al. (2011); 7 equations for tropical lowland forest trees from Basuki et al. (2009); 34 equations for temperate and tropical Australia tree species (Eamus et al. 2000, Keith et al. 2000, Snowdon et al. 2000). The balance of 81 equations were sourced from the GlobAllomeTree database, as follows. First, the online database was filtered using the Input/Output search option (http://www.globallometree.org/data/search/) to include only equations with the independent variable Diameter at Breast Height (DBH, in units of cm), and the dependent variable biomass, in units kg/tree. This returned 12335 records. These were then manually filtered using spreadsheet software to include only those equations that predicted total above-ground biomass, for trees (i.e., excluding lianes and other incompatible growth forms), and of the form specified by Eq. 1 in the main text. After accounting for duplicate equation entries (where the same equation is applied to multiple species), this left 81 unique equations that met the above criteria, spanning tropical, temperate and boreal biomes.

The results were analyzed in two ways. In the first, the model slope and intercept parameters were displayed as a scatter plot, for both the 23 models included in the analysis, and the 189 models from the global review (Fig. A1). In the second, the uncertainty around the fitted regression line was quantified as the standard deviation of the residuals, s. Estimates of s were available for all 23 equations in the current analysis and for 26 of the 189 equations from the global dataset.

Results

Fig. A1 compares the distribution of model parameters (i.e., the slope (b) and intercept (a) of the model $\ln(B) = a + b \times \ln(\text{DBH})$ for the global database and the 23 equations used in this study. Although comprising just 23 equations dominated by two genera, the morphological variability across these equations falls within the global scatter, and encompasses approximately 70% of the variability in the global data (i.e., approximately 30% of the equations from the global database have slope and/or intercept values that are greater than and/or less than the values from the 23 equations).

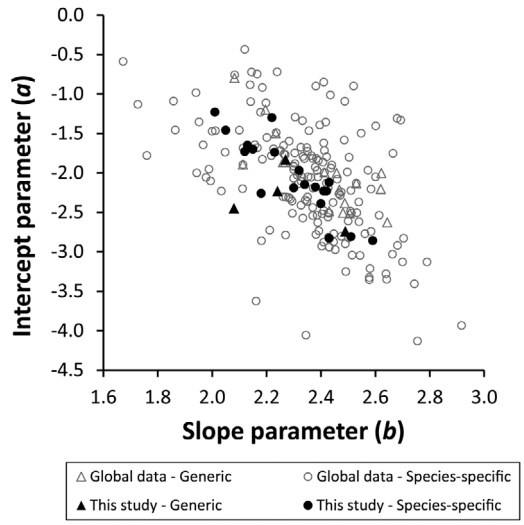

Fig. A1. Slope and intercept parameters of Eq. 1 for the 23 allometric models included in this study (Table 1), and for 189 allometric models of the same structural form obtained from the literature and from the GlobAllomeTree database (http://www.globallometree.org/).

The mean (SD) of s across the 23 datasets was 0.32 (0.07). For the global database (n = 26) the mean (SD) was 0.29 (0.10). This indicates the variability around the fitted regression model in the 23 datasets used in the main analyses is similar to the variability present in the global sample.

Taken together, these results provides confidence in the representativeness of the main analyses, and demonstrates that the variability among the 23 equations is consistent with the variability expected more widely in the allometry of trees modelled by Eq. 1.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix B

Full simulation results

Table B1. Full simulation results for Scenarios 1 and 2. Values are the required number of individuals to construct each allometric model in order to achieve the required level of precision (CV = 2.5, 5, 7.5 and 10%), for six stem diameter selection strategies.

	[S1] Quantile sampling based on inventory diameter distribution				[S2] Quantile sampling based on allometric data			
Allometric model	CV 2.5%	CV 5.0%	CV 7.5%	CV 10.0%	CV 2.5%	CV 5%	CV 7.5%	CV 10%
Mallee-form *Eucalyptus*								
E. lox-liss†	153	38	17	10	150	38	17	9
E. polybractea	219	56	25	14	211	54	24	14
E. kochii	239	60	27	15	238	60	27	15
Mean	203.9	51.4	23.0	13.0	199.8	50.3	22.5	12.7
SD	44.9	11.5	5.2	2.9	45.1	11.5	5.2	2.9
Eucalyptus and *Acacia* trees								
E. wandoo	280	71	32	18	457	117	53	30
E. viminalis	100	25	11	6	133	34	15	9
E. tricarpa-sideroxylon	175	44	19	11	166	42	19	11
E. tereticornis	107	27	12	7	155	40	18	10
E. loxophleba	106	27	12	7	105	27	12	7
E. spathulata	108	27	12	7	180	46	21	12
E. polyanthermos	113	29	13	7	144	36	16	9
E. occidentalis	98	25	11	6	191	50	23	13
E. melliodora	182	46	21	12	195	49	22	12
E. largiflorens	67	17	8	4	104	27	12	7
E. camaldulensis	127	32	14	8	134	34	15	9
A. acuminata	377	95	42	24	606	154	69	39
A. melanoxylon	114	29	13	7	115	29	13	7
A. saligna	246	61	27	15	382	98	44	25
Mean	157.2	39.6	17.7	10.0	219.1	55.9	25.1	14.3
SD	87.3	21.9	9.8	5.5	151.8	38.7	17.4	9.9
Acacia shrubs								
A. calamifolia	231	59	26	15	284	71	32	18
A. hakeoides	210	53	24	13	267	68	30	17
Mean	220.4	55.8	25.0	14.1	275.1	69.6	31.1	17.6
SD	14.6	3.8	1.8	1.0	12.0	2.5	1.0	0.5
Generalized allometrics								
Generic non-*Acacia* shrubs	297	74	33	19	305	77	35	20
Generic *Acacia* shrubs	363	91	41	23	358	90	40	22
Generic Mallee-form eucalyptus	223	57	26	15	215	54	24	14
Generic tree	226	57	25	14	250	64	29	16

† *Eucalyptus loxophleba* ssp. *lissophloia*.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B2. Full simulation results for Scenarios 3 and 4. Values are the required number of individuals to construct each allometric model in order to achieve the required level of precision (CV = 2.5, 5, 7.5 and 10%), for six stem diameter selection strategies.

Allometric model	[S3] Uniform random (natural scale)				[S4] Stratified random (natural scale)			
	CV 2.5%	CV 5.0%	CV 7.5%	CV 10.0%	CV 2.5%	CV 5%	CV 7.5%	CV 10%
Mallee-form *Eucalyptus*								
E. lox-liss†	150	38	17	10	148	37	16	9
E. polybractea	260	67	31	17	257	65	29	17
E. kochii	232	58	26	15	230	57	25	14
Mean	214.1	54.6	24.6	13.9	211.7	53.1	23.7	13.3
SD	57.3	15.0	6.9	4.0	56.7	14.5	6.5	3.7
Eucalyptus and *Acacia* trees								
E. wandoo	293	74	33	18	294	73	32	18
E. viminalis	98	25	11	6	96	24	11	6
E. tricarpa-sideroxylon	178	46	21	12	175	44	20	11
E. tereticornis	110	28	13	7	107	27	12	7
E. loxophleba	143	37	17	9	138	34	15	9
E. spathulata	112	29	13	7	108	27	12	7
E. polyanthermos	116	29	13	7	114	29	13	7
E. occidentalis	123	32	15	8	119	30	13	8
E. melliodora	306	80	37	21	296	74	33	19
E. largiflorens	101	26	12	7	100	25	11	6
E. camaldulensis	315	83	38	22	307	78	35	20
A. acuminata	381	97	44	25	381	96	43	24
A. melanoxylon	133	34	15	9	131	33	15	8
A. saligna	237	60	27	15	234	59	26	15
Mean	189.1	48.6	22.0	12.5	185.6	46.6	20.8	11.7
SD	97.2	25.2	11.4	6.5	96.7	24.3	10.8	6.1
Acacia shrubs								
A. calamifolia	235	59	27	15	232	58	26	14
A. hakeoides	205	52	23	13	204	51	23	13
Mean	220.2	55.6	24.9	14.1	217.9	54.4	24.2	13.6
SD	21.0	5.3	2.4	1.3	20.2	4.9	2.2	1.2
Generalized allometrics								
Generic non-*Acacia* shrubs	472	122	55	32	467	118	53	30
Generic *Acacia* shrubs	569	148	67	39	571	143	64	36
Generic Mallee-form eucalyptus	250	64	29	16	247	62	28	16
Generic tree	633	166	76	44	629	160	71	40

†*Eucalyptus loxophleba* ssp. *lissophloia*.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B3. Full simulation results for Scenarios 5 and 6. Values are the required number of individuals to construct each allometric model in order to achieve the required level of precision (CV = 2.5, 5, 7.5 and 10%), for six stem diameter selection strategies.

Allometric model	[S5] Uniform random (logarithmic scale)				[S6] Stratified random (logarithmic scale)			
	CV 2.5%	CV 5.0%	CV 7.5%	CV 10.0%	CV 2.5%	CV 5%	CV 7.5%	CV 10%
Mallee-form *Eucalyptus*								
E. lox-liss†	214	55	25	14	211	53	23	13
E. polybractea	250	63	28	16	250	63	28	16
E. kochii	407	103	46	26	402	100	44	25
Mean	290.3	73.8	33.1	18.7	287.8	71.9	32.0	18.0
SD	102.4	25.9	11.6	6.6	101.0	24.9	11.0	6.2
Eucalyptus and *Acacia* trees								
E. wandoo	488	124	56	32	485	122	54	31
E. viminalis	134	34	15	8	134	34	15	8
E. tricarpa-sideroxylon	191	49	22	12	189	47	21	12
E. tereticornis	133	34	15	9	129	32	14	8
E. loxophleba	103	26	12	7	102	26	11	6
E. spathulata	135	34	15	9	133	33	15	8
E. polyanthermos	188	48	22	12	185	46	21	12
E. occidentalis	118	30	13	8	116	29	13	7
E. melliodora	181	45	20	11	179	45	20	11
E. largiflorens	172	44	20	11	171	43	19	11
E. camaldulensis	125	32	14	8	122	30	13	8
A. acuminata	518	131	59	33	514	128	57	32
A. melanoxylon	133	34	15	8	131	32	14	8
A. saligna	330	84	37	21	325	81	36	20
Mean	210.6	53.4	23.9	13.5	208.2	52.1	23.2	13.0
SD	135.7	34.5	15.5	8.8	135.3	33.8	15.0	8.5
Acacia shrubs								
A. calamifolia	401	102	46	26	395	99	44	25
A. hakeoides	347	87	39	22	345	87	39	22
Mean	373.9	94.6	42.3	23.9	370.4	93.0	41.4	23.3
SD	38.2	10.5	4.9	2.9	35.3	8.7	3.9	2.2
Generalized allometrics								
Generic non-*Acacia* shrubs	293	75	34	19	292	73	32	18
Generic *Acacia* shrubs	357	91	41	23	352	88	39	22
Generic Mallee-form eucalyptus	266	67	30	17	263	67	30	17
Generic tree	249	64	29	16	245	62	27	15

†*Eucalyptus loxophleba* ssp. *lissophloia*.

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B4. Summary of Tables B1–B3. Table entries are the required number of individuals to construct allometric models to achieve 5%, 7.5% and 10% predicted biomass CV_B for four of the six stem diameter selection methods. For the species-specific allometric models (Eucalypt trees, *Acacia* trees and shrubs, and mallee-form eucalypts) the mean is given, as well as the minimum and maximum values across all species within each class (in parentheses).

	Stem diameter selection method			
CV_B critical value	S1	S2	S4	S6
Predicted biomass $CV_B = 5\%$				
Eucalypt and *Acacia* trees	40 (17, 95)	56 (27, 154)	47 (24, 96)	52 (26, 128)
Acacia shrubs	56 (53,59)	70 (68, 71)	54 (51,58)	93 (87,99)
Mallee-form eucalypts	51 (38, 60)	50 (38, 60)	53 (37, 65)	72 (53, 100)
Generic tree	57	64	160	62
Generic mallee *Eucalyptus*	57	54	64	67
Generic non-*Acacia* shrub	74	77	118	73
Generic *Acacia* shrub	91	90	143	88
Predicted biomass $CV_B = 7.5\%$				
Eucalypt and *Acacia* trees	18 (8, 42)	25 (12, 69)	21 (11, 43)	23 (11, 57)
Acacia shrubs	25 (24,26)	31 (30, 32)	24 (23,26)	41 (39, 44)
Mallee-form eucalypts	23 (17, 27)	23 (17, 27)	24 (16, 29)	32 (23, 44)
Generic tree	25	29	71	27
Generic mallee *Eucalyptus*	26	24	29	28
Generic non-*Acacia* shrub	33	35	53	30
Generic *Acacia* shrub	41	40	64	39
Predicted biomass $CV_B = 10\%$				
Eucalypt and *Acacia* trees	10 (4, 24)	14 (7, 39)	12 (6, 24)	13 (6, 32)
Acacia shrubs	15 (7,24)	18 (17, 18)	14 (13,14)	23 (22, 25)
Mallee-form eucalypts	14 (13, 15)	13 (9, 15)	13 (9, 17)	18 (13, 25)
Generic tree	14	16	40	15
Generic mallee *Eucalyptus*	15	14	16	17
Generic non-*Acacia* shrub	19	20	30	18
Generic *Acacia* shrub	23	22	36	22

Note: The stem diameter selection methods are: S1, individuals selected at random from observed field diameter distribution; S2, individuals selected at random from allometric diameter distribution; S4, individuals selected at random, uniform on the natural scale (5 strata); S6, individuals selected at random, uniform on the log scale (5 strata).

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix C

Tables of critical values

Table C1. Expected precision for biomass allometric models with individuals selected under Scenario 1 (individuals selected to ensure commonality in diameter distribution with field inventory data). The table gives the expected biomass precision (CV_B) for a given level of variability in the relationship between stem diameter and biomass (s) and sample size (n_s).

	Sample size (n_s)									
Residual SD, s	10	20	30	40	50	60	70	80	90	100
0.200	6.6	4.7	3.8	3.3	3.0	2.7	2.5	2.3	2.2	2.1
0.205	6.8	4.8	3.9	3.4	3.0	2.8	2.6	2.4	2.3	2.1
0.210	6.9	4.9	4.0	3.5	3.1	2.8	2.6	2.5	2.3	2.2
0.215	7.1	5.0	4.1	3.6	3.2	2.9	2.7	2.5	2.4	2.2
0.220	7.3	5.1	4.2	3.6	3.3	3.0	2.7	2.6	2.4	2.3
0.225	7.4	5.3	4.3	3.7	3.3	3.0	2.8	2.6	2.5	2.4
0.230	7.6	5.4	4.4	3.8	3.4	3.1	2.9	2.7	2.5	2.4
0.235	7.8	5.5	4.5	3.9	3.5	3.2	2.9	2.7	2.6	2.5
0.240	7.9	5.6	4.6	4.0	3.5	3.2	3.0	2.8	2.6	2.5
0.245	8.1	5.7	4.7	4.1	3.6	3.3	3.1	2.9	2.7	2.6
0.250	8.3	5.9	4.8	4.1	3.7	3.4	3.1	2.9	2.8	2.6
0.255	8.4	6.0	4.9	4.2	3.8	3.4	3.2	3.0	2.8	2.7
0.260	8.6	6.1	5.0	4.3	3.9	3.5	3.3	3.0	2.9	2.7
0.265	8.8	6.2	5.1	4.4	3.9	3.6	3.3	3.1	2.9	2.8
0.270	9.0	6.3	5.2	4.5	4.0	3.7	3.4	3.2	3.0	2.8
0.275	9.2	6.5	5.3	4.6	4.1	3.7	3.5	3.2	3.1	2.9
0.280	9.3	6.6	5.4	4.7	4.2	3.8	3.5	3.3	3.1	3.0
0.285	9.5	6.7	5.5	4.8	4.3	3.9	3.6	3.4	3.2	3.0
0.290	9.7	6.8	5.6	4.8	4.3	4.0	3.7	3.4	3.2	3.1
0.295	9.9	7.0	5.7	4.9	4.4	4.0	3.7	3.5	3.3	3.1
0.300	10.0	7.1	5.8	5.0	4.5	4.1	3.8	3.6	3.3	3.2
0.305	10.2	7.2	5.9	5.1	4.6	4.2	3.9	3.6	3.4	3.2
0.310	10.4	7.4	6.0	5.2	4.7	4.3	3.9	3.7	3.5	3.3
0.315	10.6	7.5	6.1	5.3	4.7	4.3	4.0	3.7	3.5	3.3
0.320	10.8	7.6	6.2	5.4	4.8	4.4	4.1	3.8	3.6	3.4
0.325	11.0	7.7	6.3	5.5	4.9	4.5	4.1	3.9	3.7	3.5
0.330	11.1	7.9	6.4	5.6	5.0	4.5	4.2	3.9	3.7	3.5
0.335	11.3	8.0	6.5	5.7	5.1	4.6	4.3	4.0	3.8	3.6
0.340	11.5	8.1	6.6	5.8	5.1	4.7	4.3	4.1	3.8	3.6
0.345	11.7	8.3	6.7	5.8	5.2	4.8	4.4	4.1	3.9	3.7
0.350	11.9	8.4	6.8	5.9	5.3	4.8	4.5	4.2	4.0	3.8
0.355	12.0	8.5	7.0	6.0	5.4	4.9	4.6	4.3	4.0	3.8
0.360	12.2	8.6	7.1	6.1	5.5	5.0	4.6	4.3	4.1	3.9
0.365	12.4	8.8	7.2	6.2	5.5	5.1	4.7	4.4	4.1	3.9
0.370	12.6	8.9	7.3	6.3	5.6	5.1	4.8	4.4	4.2	4.0
0.375	12.8	9.0	7.4	6.4	5.7	5.2	4.8	4.5	4.3	4.0
0.380	12.9	9.1	7.5	6.5	5.8	5.3	4.9	4.6	4.3	4.1
0.385	13.1	9.3	7.6	6.6	5.9	5.4	5.0	4.6	4.4	4.1
0.390	13.3	9.4	7.7	6.6	5.9	5.4	5.0	4.7	4.4	4.2
0.395	13.5	9.5	7.8	6.7	6.0	5.5	5.1	4.8	4.5	4.3
0.400	13.6	9.6	7.9	6.8	6.1	5.6	5.1	4.8	4.5	4.3
0.405	13.8	9.7	8.0	6.9	6.2	5.6	5.2	4.9	4.6	4.4
0.410	14.0	9.9	8.1	7.0	6.2	5.7	5.3	4.9	4.7	4.4
0.415	14.1	10.0	8.2	7.1	6.3	5.8	5.3	5.0	4.7	4.5
0.420	14.3	10.1	8.2	7.1	6.4	5.8	5.4	5.0	4.8	4.5
0.425	14.4	10.2	8.3	7.2	6.5	5.9	5.5	5.1	4.8	4.6
0.430	14.6	10.3	8.4	7.3	6.5	6.0	5.5	5.2	4.9	4.6
0.435	14.8	10.4	8.5	7.4	6.6	6.0	5.6	5.2	4.9	4.7
0.440	14.9	10.5	8.6	7.5	6.7	6.1	5.6	5.3	5.0	4.7
0.445	15.1	10.6	8.7	7.5	6.7	6.1	5.7	5.3	5.0	4.8
0.450	15.2	10.8	8.8	7.6	6.8	6.2	5.7	5.4	5.1	4.8
0.455	15.4	10.9	8.9	7.7	6.9	6.3	5.8	5.4	5.1	4.9
0.460	15.5	11.0	8.9	7.7	6.9	6.3	5.9	5.5	5.2	4.9
0.465	15.6	11.1	9.0	7.8	7.0	6.4	5.9	5.5	5.2	4.9

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table C1. Continued.

	Sample size (n_s)									
Residual SD, s	10	20	30	40	50	60	70	80	90	100
0.470	15.8	11.2	9.1	7.9	7.1	6.4	6.0	5.6	5.3	5.0
0.475	15.9	11.2	9.2	8.0	7.1	6.5	6.0	5.6	5.3	5.0
0.480	16.0	11.3	9.3	8.0	7.2	6.5	6.1	5.7	5.3	5.1
0.485	16.2	11.4	9.3	8.1	7.2	6.6	6.1	5.7	5.4	5.1
0.490	16.3	11.5	9.4	8.1	7.3	6.6	6.2	5.8	5.4	5.1
0.495	16.4	11.6	9.5	8.2	7.3	6.7	6.2	5.8	5.5	5.2
0.500	16.5	11.7	9.5	8.3	7.4	6.7	6.2	5.8	5.5	5.2

Table C2. Expected required sample sizes for biomass allometric models with individuals selected under Scenario 1 (individuals selected to ensure commonality in diameter distribution with field inventory data). The table gives the required sample sizes to achieve a given level of precision in estimated biomass (CV_B), for an allometric model with a given value of s.

	Precision (CV_B)				
Residual SD, s	1	2.5	5	7.5	10
0.200	440	70	18	8	4
0.205	461	74	18	8	5
0.210	483	77	19	9	5
0.215	505	81	20	9	5
0.220	529	85	21	9	5
0.225	553	88	22	10	6
0.230	578	92	23	10	6
0.235	603	97	24	11	6
0.240	630	101	25	11	6
0.245	657	105	26	12	7
0.250	685	110	27	12	7
0.255	714	114	29	13	7
0.260	744	119	30	13	7
0.265	774	124	31	14	8
0.270	805	129	32	14	8
0.275	837	134	33	15	8
0.280	870	139	35	15	9
0.285	904	145	36	16	9
0.290	938	150	38	17	9
0.295	974	156	39	17	10
0.300	1010	162	40	18	10
0.305	1046	167	42	19	10
0.310	1084	173	43	19	11
0.315	1122	180	45	20	11
0.320	1161	186	46	21	12

Table C2. Continued.

	Precision (CV_B)				
Residual SD, s	1	2.5	5	7.5	10
0.325	1200	192	48	21	12
0.330	1240	198	50	22	12
0.335	1281	205	51	23	13
0.340	1323	212	53	24	13
0.345	1365	218	55	24	14
0.350	1407	225	56	25	14
0.355	1450	232	58	26	15
0.360	1494	239	60	27	15
0.365	1538	246	62	27	15
0.370	1582	253	63	28	16
0.375	1627	260	65	29	16
0.380	1672	268	67	30	17
0.385	1718	275	69	31	17
0.390	1763	282	71	31	18
0.395	1809	289	72	32	18
0.400	1855	297	74	33	19
0.405	1901	304	76	34	19
0.410	1947	312	78	35	19
0.415	1994	319	80	35	20
0.420	2040	326	82	36	20
0.425	2086	334	83	37	21
0.430	2132	341	85	38	21
0.435	2177	348	87	39	22
0.440	2223	356	89	40	22
0.445	2268	363	91	40	23
0.450	2313	370	93	41	23
0.455	2357	377	94	42	24
0.460	2401	384	96	43	24
0.465	2444	391	98	43	24
0.470	2487	398	99	44	25
0.475	2529	405	101	45	25
0.480	2570	411	103	46	26
0.485	2611	418	104	46	26
0.490	2651	424	106	47	27
0.495	2689	430	108	48	27

21508925, 2015, 3, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/ES14-00251.1, Wiley Online Library on [06/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License